

Digital Mice: Inducing and Validating Controlled Cognitive Deficits in Frontier Language Models for Mechanism Design

May 28, 2026

Abstract

Early-stage mechanism design is a phase-I science: the relevant behavior is human cognition, so designers must run experiments on people before they can learn which mechanisms are robust to bounded rationality. We ask whether frontier reasoning models can serve as the economist’s analogue of the laboratory mouse—capable reasoners instructed into specific, controlled cognitive deficits whose behavior can be studied before human subjects are recruited. Using a canonical second-price sealed-bid auction (SPSB) and a $3 \times 3 \times 3$ panel of *digital mice* defined over three cognitive axes (contingent reasoning, forward planning, belief reasoning), we establish three facts. First, the deficits have *construct validity*: repairing each axis moves truthful bidding in the theory-predicted direction, monotonically and with the predicted sign in all five pre-registered directional tests. Second, the panel—augmented by a small set of targeted error strains—attains *calibration coverage*: it reproduces all five canonical SPSB human-mistake classes (truthful, moderate underbidding, overbidding, extreme overbidding, and near-zero bidding) at rates above their detection thresholds. Third, the panel is *discriminating*: in a first-price sealed-bid stress test the reasoning strains track the risk-neutral Bayes–Nash benchmark ($b = \frac{2}{3}v$), overbid relative to it at the rate documented for human subjects, and a strain that overgeneralizes the second-price payment rule overbids above value and loses money—a mechanistic account of a winner’s-curse pathology. We also show that augmenting the deficit instruction with a worked chain-of-thought demonstration sharpens and stabilizes the induced behavior, and that at the strain level four of five reserved deficits preserve their behavioral class across base models. Finally, when we let the panel drive an evolutionary search over a structured genotype space of

single-item auctions (24 individuals, 2 generations, 480 raw bidder-decisions), the search improves the best fitness across generations and concentrates the top of the population on canonical sealed-bid designs, with first-price continuous emerging on top—the same mechanism the a-priori stress test selected. These results support the use of instructed-deficit language models as a phase-I screening population for mechanism design.

Keywords: bounded rationality, mechanism design, second-price auctions, large language models, construct validity.

1 Introduction

Medicine studies humans, but it does not begin with them. Much early drug testing starts with mice, and some of those mice are deliberately engineered to lack a specific protein, receptor, or channel. The deficit is not a nuisance; it is the identification strategy. If a treatment succeeds or fails only when a particular channel is removed, the experiment isolates a mechanism that is entangled in an intact organism.

Economics has lacked this luxury. The behavior of interest in mechanism design is human cognition—contingent reasoning, forward planning, belief formation, and strategic sophistication—and these channels are bundled in real subjects. A bidder who fails to bid truthfully in a strategy-proof auction may lack contingent reasoning, may fail to plan forward, may hold incorrect beliefs about rivals, or may simply misunderstand the rule. Natural and laboratory experiments rarely deliver a population missing exactly one cognitive module, so the empirical content of theories such as obvious strategy-proofness (Li, 2017), cursed equilibrium (Eyster and Rabin, 2005), and level- k reasoning (Crawford and Iriberri, 2007) is hard to separate.

Large language models change the experimental object. The strongest current reasoning models are, in simple games, often closer to a fully rational agent than an ordinary subject: they follow long chains of reasoning and compute through cases. That capability is precisely what makes them useful here. We start from a capable reasoning substrate and instruct it to “play dumb” in a controlled way—reason only myopically, enumerate only local contingencies, ignore second-order beliefs—through explicit procedural constraints on its deliberation, not through dependence on hidden quirks of one model. This yields *digital mice*: language-model agents with specific, engineered cognitive deficits.

Research question. Can a frontier reasoning model be instructed into stable, separable bounded-rationality states that (i) produce the behavioral signatures predicted by cognitive theory and (ii) span the empirically documented space of human mistakes, in a canonical second-price sealed-bid auction?

Contribution. We answer both halves affirmatively for the SPSB environment. Over a $3 \times 3 \times 3$ panel of 27 strains (810 bidder-decisions), repairing each cognitive axis raises the truthful-bid rate by +0.23 (contingent reasoning), +0.14 (forward planning), and +0.05 (belief reasoning); all five pre-registered directional hypotheses pass their sign test. The panel, augmented by two targeted error strains, covers all five canonical SPSB human-mistake classes at rates above threshold. Third, in a first-price stress test the reasoning strains shade to the risk-neutral BNE ($b = \frac{2}{3}v$, median ratios 0.65–0.67) and overbid relative to it at 38%, while the second-price-overgeneralizer mouse bids above value and loses money. Along the way we show that a worked chain-of-thought demonstration is a stronger control than instruction alone—it drives the all-deficit strain’s truthful rate from 0.39 to 0 and the all-repaired strain’s to a perfect 1.0 while lowering bid dispersion—and that four of five reserved deficits keep their behavioral class across base models. As a last check, an evolutionary search that uses F_{robust} on the digital-mouse panel as its fitness over a structured single-item-auction genotype space independently elevates first-price sealed-bid designs to the top of the population, agreeing with the a-priori choice. We are explicit about a boundary of the construction: the clean procedural basis naturally produces truthful and underbidding behavior but not overbidding, which we recover with a small number of hand-specified heuristic strains. We treat the digital-mouse panel as phase-I screening evidence, not as a prediction of human population shares.

2 The Digital-Mouse Panel

2.1 Construction: a rational base plus a deficit instruction

A digital mouse is a single prompt assembled from two layers. The first layer, the *rational base*, is identical across all strains and instructs the model to play the auction as a long-run profit maximizer:

*Your TOP PRIORITY is to place bids which maximize your profit in the long run.
Learn from the history of previous rounds in order to maximize your total profit.
Don’t forget the values are redrawn independently each round.*

Table 1: The $3 \times 3 \times 3$ digital-mouse panel. Each strain is one combination of axis levels; the lowest level is the engineered deficit.

Axis	Level 0 (deficit)	Level 1	Level 2 (repaired)
Contingent reasoning	no enumeration	enumerate cases	worst-case / dominance
Forward planning	myopic	one-step lookahead	tree / multi-step
Belief reasoning	no model of others	first-order beliefs	second-order beliefs

The second layer, *defect induction*, appends a “cognitive style” block containing exactly one paragraph per cognitive axis. Each paragraph constrains the model’s deliberation *procedure*—not its objective—at one of three levels. The intervention is therefore a reasoning-procedure control rather than a payoff or preference change. Concretely, the contingent-reasoning axis ranges from a level-0 paragraph that forbids reasoning about opponents to a level-2 paragraph that mandates a worst-case/dominance check:

Contingent reasoning, level 0 (deficit). *Do NOT enumerate possible bids by other bidders or compute best responses to specific opponent moves. Choose your bid without conditioning on what others might do.*

Contingent reasoning, level 2 (repaired). *For each possible bid you could make, consider the worst-case outcome given the range of possible opponent bids. Find the bid where even the worst-case outcome is acceptable. If you deviated from this bid, could the worst case become worse?*

The forward-planning and belief-reasoning axes are constructed the same way (from “do not trace consequences” to a multi-step decision tree, and from “do not model others” to explicit second-order beliefs). A strain is thus the rational base plus a triple $(c, f, b) \in \{0, 1, 2\}^3$ that selects one paragraph for each axis. The all-off triple $(0, 0, 0)$ is the most deficient mouse; the all-repaired triple $(2, 2, 2)$ is closest to the rational base.

2.2 The $3 \times 3 \times 3$ grid

Varying three axes at three levels each gives the 27-strain panel of Table 1. A useful intuition: the rational base supplies the *goal* (maximize profit), and the cognitive-style block rations the *reasoning steps* the mouse is allowed to use to pursue it. A mouse told not to enumerate opponents, not to trace payoffs, and not to model others must fall back on a crude own-value heuristic; a mouse granted all three faculties can in principle reconstruct the dominant strategy.

Hardening the base (strict variant). The base layer above makes profit maximization the top priority, which leaves room for a very capable model to “see through” a deficit and revert to the optimal strategy. For the cross-model analysis of Section 6 we therefore also build a *strict* variant whose base layer inverts the priority—“follow the cognitive procedure exactly . . . if the cognitive procedure conflicts with the profit-maximizing strategy, the cognitive procedure wins”—and explicitly forbids invoking dominant-strategy or Vickrey shortcuts. The strict variant is used only where noted; all construct-validation and calibration results use the standard panel.

3 Experimental Design and Methodology

Environment. The mechanism is a single-item second-price sealed-bid auction with $N = 3$ bidders. Private values are drawn independently and uniformly on $[0, 50]$; each bidder observes only its own value and submits one sealed bid on a \$0.1 grid. Under standard preferences truthful bidding is a weakly dominant strategy, which makes truthfulness a clean construct-validity target: a more capable mouse should bid closer to its value.

Strains and runs. We instantiate all 27 basis strains with the same base model (gpt-5.4-mini, temperature 0.5) and collect 30 bidder-decisions per strain (810 in total). Because the clean procedural basis does not naturally generate *overbidding*, we additionally author a small set of targeted *error mice*—heuristic strains designed to express one documented mistake each (e.g. a “second-price overgeneralizer” that treats bidding above value as safe because the winner pays the runner-up bid, and a “payment panic” strain that bids near zero). Seven error strains are specified; two had completed runs at the time of this analysis and feed the calibration scorecard below.

Behavioral classes. For each bidder-decision we record the bid, the value, and the deviation $b - v$. We define a truthful bid as $|b - v| \leq 0.05$, moderate underbidding as $b < v - 0.05$ (excluding near-zero bids), overbidding as $b > v + 0.05$, extreme overbidding as $b - v \geq 5$, and near-zero bidding as $b \leq 0.1$ with $v > 5$. These thresholds operationalize the five canonical SPSB mistake classes used in the calibration step.

3.1 Outcome variables and the robustness objective

We now define the panel-based objective used to score mechanisms in Section 6. Fix a mechanism and a persona panel $\mathcal{P} = \{p_1, \dots, p_M\}$, and run R independent auctions per

persona. Index the resulting completed auctions by $a \in \mathcal{A}$, with $\mathcal{A}_p$ the subset run under persona p . Each auction has $N = 3$ bidders with private values $v_{a,i}$ and elicited bids $b_{a,i}$; the winner is $i^*(a) = \arg \max_i b_{a,i}$ and the mechanism's pricing rule sets the payment $\rho(a)$ (the second-highest bid under SPSB, the own bid under all-pay, and so on). Let A_{att} be the number of *attempted* auctions and F the number that fail to parse or violate the response format, so the failure rate is $\phi = F/A_{\text{att}}$.

Per-auction quantities. Write $v_a^{\max} = \max_i v_{a,i}$. We use allocative efficiency, a bid-shading dispersion, and realized winner surplus:

$$\text{eff}(a) = \frac{v_{a,i^*(a)}}{v_a^{\max}}, \quad s(a) = \text{std}_{i: v_{a,i} > 0} \left(\frac{b_{a,i}}{v_{a,i}} \right), \quad \pi(a) = v_{a,i^*(a)} - \rho(a). \quad (1)$$

Panel aggregates. With $A = |\mathcal{A}|$ and a value scale $\bar{v} = \max_a v_a^{\max}$, define normalized revenue, mean efficiency, a simplicity proxy, and normalized surplus variance:

$$\hat{\rho} = \frac{1}{A\bar{v}} \sum_a \rho(a), \quad \overline{\text{eff}} = \frac{1}{A} \sum_a \text{eff}(a), \quad \sigma = 1 - \frac{1}{A} \sum_a s(a), \quad \widehat{\text{Var}} = \frac{\text{Var}_a(\pi(a))}{\bar{v}^2}. \quad (2)$$

The simplicity proxy σ is high when bidders converge on a common bid/value ratio (an “obvious” mechanism) and low when their shading scatters.

Intended-action rate. Each mechanism declares the action a correctly reasoning bidder should take, $\tau \in \{\text{TRUTHFUL}, \text{SHADE}, \text{AGGRESSIVE}, \dots\}$, with tolerance $\epsilon = 0.1$. For a scoreable decision (v, b) with $v > 0$ define the match indicator

$$m_\tau(v, b) = \begin{cases} \mathbf{1}[|b - v| \leq \epsilon] & \tau = \text{TRUTHFUL}, \\ \mathbf{1}[\kappa < b < v - \epsilon] & \tau = \text{SHADE}, \\ \mathbf{1}[b + \epsilon \geq \alpha v] & \tau = \text{AGGRESSIVE}, \end{cases} \quad (3)$$

with floor κ and aggressiveness threshold $\alpha = 0.4$ by default. The intended-action rate μ_τ is the mean of m_τ over scoreable decisions; targets labeled UNKNOWN or MECHANISM-SPECIFIC are not scored (μ_τ omitted).

Lower-tail (per-persona) statistics. Let $\overline{\text{eff}}_p$, σ_p , and $\text{Var}_p(\pi)$ be the above computed within persona p . Robustness rewards good *worst-case* behavior across the panel, not just

the average:

$$E_{\min} = \min_p \overline{\text{eff}}_p, \quad E_{20} = Q_{0.2}(\{\overline{\text{eff}}_p\}), \quad \Sigma_{\min} = \min_p \sigma_p, \quad \hat{V}_{\max} = \frac{\max_p \text{Var}_p(\pi)}{\bar{v}^2}, \quad (4)$$

where $Q_{0.2}$ is the 20th percentile across personas.

Fitness. The aggregate and robust fitness are weighted sums

$$F_{\text{agg}} = w_1 \hat{\rho} + w_2 \overline{\text{eff}} + w_3 \mu_\tau + w_4 \sigma - w_5 \widehat{\text{Var}} - w_6 \phi, \quad (5)$$

$$F_{\text{tail}} = w_7 E_{\min} + w_8 E_{20} + w_9 \Sigma_{\min} - w_{10} \hat{V}_{\max}, \quad (6)$$

$$F_{\text{robust}} = F_{\text{agg}} + F_{\text{tail}}. \quad (7)$$

Following the Stage-C calibration mapping, the default weights are $(w_1, \dots, w_{10}) = (1, 1, 0.5, 0.5, 0.25, 1, 0.5, 0.5, 0.25, 0.25)$, grouping into domain performance (w_1, w_2) , intended action (w_3) , simplicity (w_4, w_9) , regret (w_5, w_{10}) , compliance (w_6) , and cognitive fragility (w_7, w_8) . Sections 6, 7, and 8 report F_{robust} and μ_τ ; the sign convention is that fragility, regret, and failure enter negatively, so a mechanism scores well only if it is allocatively efficient, induces its intended action, and does so *uniformly* across the digital-mouse panel. The full machinery of evolutionary search over auction designs is deferred to Section 8, which is self-contained.

4 Result 1: Construct Validity

If the cognitive axes are genuine bounded-rationality controls rather than arbitrary prompt artifacts, then repairing an axis should move behavior toward the rational benchmark. Figure 1 plots the truthful-bid rate at each axis level, marginalizing over the other two axes. All three axes are monotone or near-monotone in the repairing direction, and the contingent-reasoning axis produces the largest swing.

Table 2 reports the five pre-registered directional tests. For each axis we test that repairing it raises the truthful rate (expected positive) and, where applicable, lowers the underbidding rate (expected negative). All five tests pass: the realized sign matches the prediction and exceeds the minimum-magnitude bar in every case. The directional structure is exactly what theory predicts—more contingent reasoning and more forward planning push the agent toward its dominant strategy—which is the core construct-validity claim.

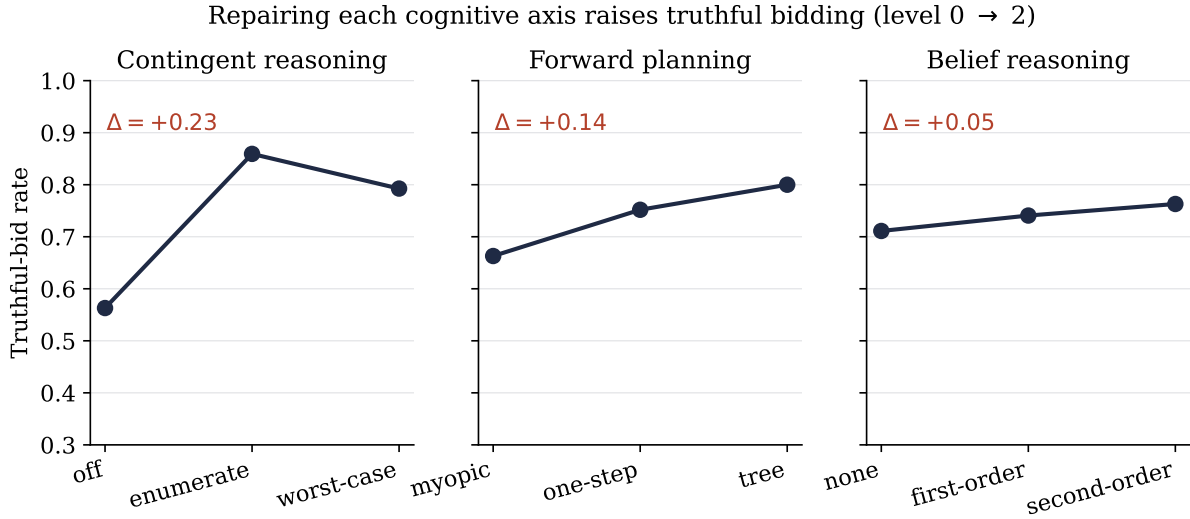


Figure 1: Construct-validity marginals. Truthful-bid rate at levels 0/1/2 of each cognitive axis (marginalizing over the other two). Δ is the level-2-minus-level-0 change. Repairing each axis raises truthfulness, with the largest effect on the contingent-reasoning axis.

Table 2: Construct-validity diagnostics: five pre-registered directional tests on the cognitive axes. Δ is the level-2-minus-level-0 change in the metric. All tests pass.

Axis	Metric	Δ (L2–L0)	$ \Delta $ bar	Expected sign	Pass
Contingent reasoning	truthful rate	+0.230	0.05	positive	✓
Contingent reasoning	underbid rate	−0.230	0.05	negative	✓
Forward planning	truthful rate	+0.137	0.05	positive	✓
Forward planning	underbid rate	−0.137	0.05	negative	✓
Belief reasoning	truthful rate	+0.052	0.03	positive	✓

Notes. Sign rate = 1.00 (5 of 5). Underlying level means (truthful rate): contingent 0.56 $\rightarrow$ 0.86 $\rightarrow$ 0.79; forward 0.66 $\rightarrow$ 0.75 $\rightarrow$ 0.80; beliefs 0.71 $\rightarrow$ 0.74 $\rightarrow$ 0.76.

Figure 2 shows that the strains are also *separable* at the level of raw behavior. The rational mouse (contingent reasoning engaged) bids on the 45° truthful line; the all-deficits-off mouse systematically underbids; and the targeted error mice occupy distinct regions—persistent overbidding above the line and near-zero bidding along the floor. The mean absolute deviation falls from 2.78 for the most deficient basis strain to essentially zero (0.003) for the most rational one.

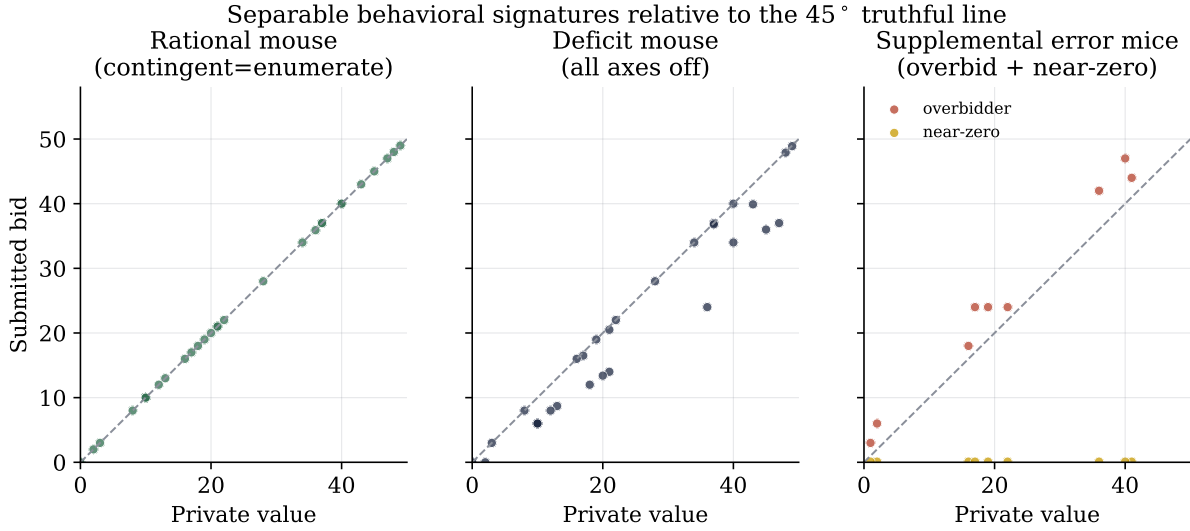


Figure 2: Separable behavioral signatures. Bid versus private value for a rational basis strain, the all-deficits-off strain, and the two supplemental error mice. The dashed line is the 45° truthful benchmark.

5 Result 2: Calibration Coverage

Construct validity establishes that the axes do what theory says. Calibration asks a different question: does the panel *span* the space of mistakes that real people make in this mechanism? We score five canonical SPSB human-mistake classes and ask, for each, whether some strain in the panel reproduces it at a rate above a detection threshold.

Figure 3 and Table 3 report the scorecard. All five classes are covered. Truthful and overbidding behavior are reproduced at very high rates (0.97 and 1.00); moderate underbidding, extreme overbidding, and near-zero bidding are reproduced at plausible intermediate rates (0.63, 0.44, 0.78). The best strain for each class is reported in the table: truthfulness and underbidding are covered by the clean procedural basis, whereas the overbidding and near-zero classes are covered by the targeted error mice—making explicit which classes the procedural construction reaches on its own and which require hand-specified heuristics.

6 Result 3: Demonstrations Sharpen Induction

The panel so far induces a deficit by *instruction*. A natural alternative, familiar from few-shot chain-of-thought prompting, is to also *demonstrate* the bounded procedure with a worked example. We append to a persona a single worked example that walks

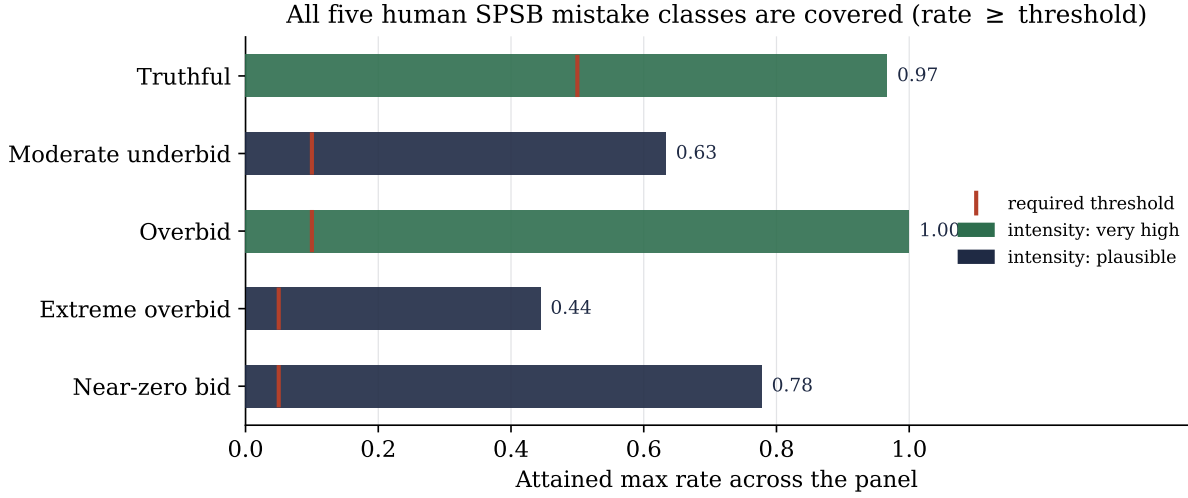


Figure 3: Calibration coverage. For each human-mistake class, the maximum rate attained by any strain in the panel (bar) against the required detection threshold (red marker). Bar color encodes the intensity band. All five classes clear their threshold.

Table 3: Calibration scorecard for the five canonical SPSB mistake classes. “Max rate” is the highest rate attained by any strain; “best strain” is the strain attaining it. All classes are covered.

Mistake class	Metric	Thresh.	Max rate	Best strain	Intensity	Covered
Truthful	truthful rate	0.50	0.967	c1_f0_b0	very high	✓
Moderate underbid	moderate underbid rate	0.10	0.633	c0_f0_b0	plausible	✓
Overbid	overbid rate	0.10	1.000	overgeneralizer [†]	very high	✓
Extreme overbid	extreme overbid rate	0.05	0.444	overgeneralizer [†]	plausible	✓
Near-zero bid	near-zero bid rate	0.05	0.778	payment panic [†]	plausible	✓

Notes. Coverage rate = 1.00 (5 of 5). [†] Supplemental error strain (spsb_error_second_price_overgeneralizer, spsb_error_payment_panic_near_zero); all other best strains are members of the clean 27-cell procedural basis.

through the allowed reasoning on a sample value and ends in a bid: a margin-of-safety underbid for the all-deficit strain, and the contingent/forward/belief reasoning that yields a truthful Vickrey bid for the all-repaired strain. We then compare instruction-only and demonstration-augmented personas on *identical* second-price value draws.

Table 4 shows that demonstrations make induction both sharper and more consistent. For the all-deficit strain, adding the worked example pulls the mean bid/value ratio down toward the demonstrated ≈ 0.7 and eliminates truthful bidding entirely (truthful rate $0.39 \rightarrow 0.00$) while *lowering* the dispersion of bid/value ratios (sd $0.124 \rightarrow 0.081$). For the all-repaired strain, the truthful demonstration produces exact Vickrey truth-telling on every decision (truthful rate $0.83 \rightarrow 1.00$, dispersion $0.076 \rightarrow 0.000$). Demonstration

Table 4: Instruction-only versus chain-of-thought (CoT) demonstration, on identical second-price draws ($n = 18$ bidder-decisions per cell). Demonstration shifts behavior toward the demonstrated procedure and lowers the dispersion of bid/value ratios.

Strain	Construction	mean b/v	truthful rate	sd(b/v)
c0_f0_b0 (all-deficit)	instruction only	0.89	0.39	0.124
c0_f0_b0 (all-deficit)	+ CoT demonstration	0.77	0.00	0.081
c2_f2_b2 (all-repaired)	instruction only	0.98	0.83	0.076
c2_f2_b2 (all-repaired)	+ CoT demonstration	1.00	1.00	0.000

is thus a stronger control than instruction: it lowers behavioral variance and makes the mouse follow the prescribed procedure more faithfully—a promising lever for the portability problem documented below, since a more forcefully induced deficit should be harder for a capable model to “repair” away.

7 Result 4: A First-Price Stress Test and Cross-Model Portability

A panel is useful for screening only if it reproduces known mechanism regularities and if its verdicts are not an artifact of the base model. We test both. For the first, we move from the second-price environment to a *first-price* sealed-bid auction, where theory and human data are sharp: with $n = 3$ risk-neutral bidders and uniform values, the symmetric Bayes–Nash equilibrium is to shade to $b = \frac{n-1}{n}v = \frac{2}{3}v$, and the canonical experimental regularity is overbidding *relative* to this risk-neutral benchmark (Kagel and Levin, 1993).

Figure 4 and Table 5 report the panel’s first-price behavior. The three reasoning strains track the equilibrium closely: their median bid/value ratios are 0.65–0.67, essentially on the $\frac{2}{3}$ BNE line, and across all reasoning decisions the mean ratio is 0.67 with 38% of bids above the risk-neutral benchmark—the documented overbidding regularity. The two error mice separate cleanly and *mechanistically*: the payment-panic mouse bids near zero, while the second-price-overgeneralizer mouse bids *above its value* (median ratio 1.15), because it applies the second-price intuition “I will not pay my own bid” to a mechanism where the winner does pay their own bid. In a first-price auction that error is not harmless—it yields negative surplus—giving a concrete, mechanism-level account of a winner’s-curse pathology.

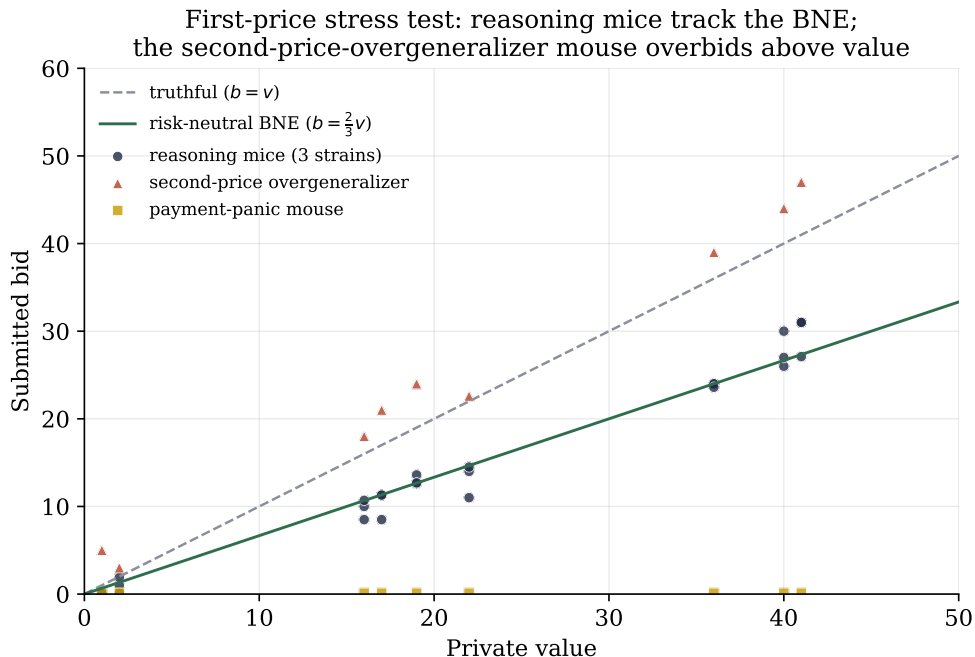


Figure 4: First-price stress test. Submitted bid versus private value for the panel. Reasoning strains cluster on the risk-neutral BNE line $b = \frac{2}{3}v$; the second-price-overgeneralizer mouse bids above the 45° value line (loss-making overbidding); the payment-panic mouse bids near zero.

Strain-level portability. The first-price result raises a prior question: do the deficit *strains* themselves behave consistently across models? A pilot exposed a failure mode—the stronger model spontaneously “repaired” the most deficient basis strain into truthful Vickrey bidding—so we hardened the panel into a *strict* scaffold that states the reasoning-procedure constraint explicitly rather than as a secondary goal. We then re-ran five reserved strict strains under both base models and compared their dominant behavioral class (Table 6). Four of five strains preserve their dominant class across models—a partial pass. The exception is the strain isolating a contingent-reasoning deficit, whose moderate-underbidding signature on `gpt-5.4-mini` is repaired into truthful bidding on `gpt-5.4` (truthful rate $0.33 \rightarrow 1.00$). Consistent with the construct-validation finding that contingent reasoning is the highest-leverage axis (Table 2), it is also the hardest deficit to hold fixed against a more capable model: the strict scaffold mitigates the portability problem but does not fully close it.

Table 5: First-price bid/value ratios by strain ($n = 9$ bidder-decisions each). The risk-neutral BNE benchmark is $\frac{2}{3} \approx 0.667$. Reasoning strains sit on the benchmark; the error mice separate above value and near zero.

Strain	Role	mean b/v	median b/v
c0_f0_b0	reasoning (all-deficit)	0.61	0.65
c2_f0_b2	reasoning (forward-off)	0.61	0.67
c2_f2_b2	reasoning (all-repaired)	0.52	0.66
overgeneralizer	error: overbid	1.61	1.15
payment panic	error: near-zero	0.02	0.01

Notes. Pooled over reasoning strains, the mean ratio is 0.67 (the BNE), 38% of bids exceed the benchmark, and 20% exceed value. The overgeneralizer’s above-value bids are loss-making under first-price pricing.

Table 6: Strain-level cross-model portability on the strict panel. Dominant behavioral class of five reserved strict strains under each base model. Four of five agree; the contingent-isolated strain’s deficit is repaired away by the stronger model.

Strain	Isolated role	mini $\rightarrow$ gpt-5.4 (dominant class)	Agree
c2_f2_b0	belief deficit	truthful $\rightarrow$ truthful	✓
c2_f0_b2	forward deficit	mod. underbid $\rightarrow$ mod. underbid	✓
c0_f0_b1	near-all-defect	mod. underbid $\rightarrow$ mod. underbid	✓
c1_f1_b1	all-intermediate	truthful $\rightarrow$ truthful	✓
c0_f2_b2	contingent deficit	mod. underbid $\rightarrow$ truthful	—

Notes. Agreement = 4/5. “Isolated role” names the axis held in its deficit state (others repaired), or a corner strain. The contingent-deficit strain’s truthful rate rises $0.33 \rightarrow 1.00$ across models; all other strains’ dominant class is stable.

8 Evolutionary Search over Auction Mechanisms

So far the panel has been used to *verify* mechanisms picked by hand (Sections 6 and 7). The natural next question is whether the same panel can *drive* a search over mechanisms. This section makes that loop concrete. The audience for what follows is the reader who has not previously seen an evolutionary algorithm; we lay out the procedure, why language models change what is possible, our specific setting, and the result.

8.1 Why an evolutionary algorithm?

Mechanism design is, at bottom, a search problem: among the auctions one could implement, which ones perform well? The economist’s standard answer is analytic—Myerson’s optimal auction, Vickrey–Clarke–Groves, OSP variants (Li, 2017; Vickrey, 1961)—each derived in closed form from a tight model of rational bidders and a known value distribution.

When we replace the rational benchmark with a panel of boundedly rational digital mice (Section 2), the closed-form route no longer applies: “best for the strict 5-mouse panel” is not a function one can differentiate or solve in closed form.

An *evolutionary algorithm* (EA) is a heuristic search procedure suited to exactly this situation. It does not require gradients, convexity, or a parameterized family that admits first-order conditions; it only needs the ability to *evaluate* a candidate against an objective. The procedure is the following loop. (i) Maintain a population of N candidate mechanisms. (ii) Score each candidate by a fitness function F . (iii) Retain the top performers (the *elites*). (iv) Produce a new generation of size N by applying small perturbations to elites (*mutation*) and recombinations of pairs of elites (*crossover*). (v) Iterate. Over generations, the population concentrates on regions of the design space where F is high.

EAs are routine in machine learning and engineering but largely absent from economic mechanism design, where searches are usually done with pencils. The reason has been operational: previous EAs over auction mechanisms have had to specify candidates as numeric vectors in a fixed family (e.g. “set the reserve price, set the entry fee”) and have had to simulate bidders with a closed-form behavioral model, typically risk-neutral BNE. Neither restriction is comfortable: many design choices in real mechanisms live in the rule text rather than in numeric parameters, and many of the empirical pathologies of real bidders are exactly the behaviors a rational benchmark cannot generate.

8.2 How language models change the picture

Frontier language models loosen both restrictions. On the bidder side, the digital-mouse panel of Section 2 supplies instructable, cognitively varied bidder agents whose deficits are procedural controls rather than implicit artifacts of one closed-form benchmark. On the mechanism side, an LLM can map a structured *genotype*—a JSON record listing payment rule, bid language, reserve, framing—into a coherent natural-language rule that the bidder LLMs can act on. The genotype carries the discrete design choices that the EA operates on; the rule text carries the implementation that the bidders see.

This decoupling is the key methodological move. Mutation and crossover operate at the *genotype* level, where they are well-typed and schema-validated: “flip `bid_language` from `continuous` to `discrete_grid`” is an unambiguous operation on a JSON record. The bidder-facing artifact, however, remains language: the LLM compiles the perturbed genotype into a rule the bidders can read. The result is an EA that can search over a design space which is genuinely discrete and categorical, on a bidder population whose behavior captures bounded rationality directly. To our knowledge, this combination—

structured-genotype EA plus instructable bounded-rational bidders plus a robustness fitness aggregated across a cognitive-deficit panel—is new to mechanism design.

What “fitness” means here. The fitness function F_{robust} of equation (7) is panel-based: a candidate mechanism scores well only if it is allocatively efficient, induces its declared intended action, and does so *across* the digital-mouse panel, not just for a rational outlier. The search is therefore explicitly searching for mechanisms that are robust to bounded rationality, not for mechanisms that are optimal against a rational benchmark. This is why the EA-plus-panel loop is the natural pairing for the research program of this paper.

8.3 Our setting

The genotype. A candidate is a JSON record (`single_item_auction_genotype_v1`) with five blocks. The *environment* fixes the value model (IPV with private range and increment), the number of bidders, and any feedback channel. The *mechanism* block sets the interaction form (sealed-bid here), the allocation rule (highest-bid wins), and—most consequentially—the payment rule (one of first-price, second-price, posted-price, all-pay, etc.), the bid language (continuous, discrete-grid, binary-accept, etc.), the reserve rule, and the tie-breaking. The *framing* block carries presentation choices that do not change the rule (menu vs. prose description, hint scaffolds, whether to include a worked chain-of-thought demonstration in the style of Section 6). A *target_policy* block declares the action a correctly reasoning bidder should take, including a benchmark formula (e.g. $b = \frac{n-1}{n}v$) and a tolerance, which feeds the intended-action match indicator m_τ of equation (3). A *complexity* block records simple syntactic features used for diagnostics. A concrete top genotype reads:

```
environment:
  value_model:  IPV, n_bidders:  3,
  private_range: [0, 49], increment: 0.1
mechanism:
  payment_rule:  first_price,
  bid_language:  continuous_bid,
  reserve_rule:  none, tie_breaking: random
framing:
  description_style: menu, scaffold: none,
  cot_demo:  false
target_policy:
  benchmark:  ((n-1)/n)*value,
```

```
target_bid_value_ratio: 0.667,
tolerance: 0.1
```

Operators. A *genotype mutation* samples a small number of fields in g and resamples their values from the allowed enumeration, then asks the LLM to rewrite the rule text consistently with the perturbed genotype—e.g. flip `bid_language` from `continuous` to `discrete_grid` and replace “submit a bid in \$0.1 increments” with “choose a bid from the grid $\{0, 1, 5, 10, 20, 30, 50\}$ ”. A *genotype crossover* takes two parents and, for each field independently, draws the child’s value from one parent with probability $\frac{1}{2}$, then again invokes the LLM rewrite. A schema validator (Section 3.1) rejects ill-typed offspring; duplicates by rule-text hash are discarded.

Selection and panel. Each generation we score every candidate by F_{robust} (eq. 7) against the *strict* 5-mouse panel of Section 7, with 2 repetitions per mouse. We retain the top quartile as elites and fill the remaining 20 slots with 50/50 mutation and crossover offspring of tournament-selected parents (tournament size 3).

Scale. The run reported below uses 24 individuals for 2 generations against the strict 5-mouse panel with 2 repetitions per mouse, for a total of 48 mechanism evaluations and 480 raw bidder-decisions, under the base model `gpt-5.4-mini`. This is a proof-of-concept of the EA-plus-panel loop; a paper-scale run would extend the horizon and re-score survivors on a held-out model along the lines of Section 7.

8.4 Result: the search recovers first-price

Figure 5 and Table 7 report the run. Two findings stand out. First, the search *improves*: the best-of-generation fitness rises from 3.37 (a gen-0 initialization) to 3.42 (a gen-1 crossover child), and that child remains the elite through gen-2. The absolute improvement is small over two generations, but it is a real one: the population’s best moves beyond the seeded population. Second, and more substantively, where the top of the population concentrates is the result: three of the top five evolved mechanisms are first-price sealed-bid auctions (continuous or discrete-grid), one is second-price continuous, and one is a posted-price binary-accept design. All five share no reserve and random tie-breaking. The EA, given a permissive genotype space and the digital-mouse panel as evaluator, independently converges on canonical sealed-bid auctions and elevates first-price continuous—exactly the mechanism Section 7 picked *a priori*—to the top.

Evolutionary search over auction genotypes (24 individuals $\times$ 2 generations $\times$ 5-mouse panel)

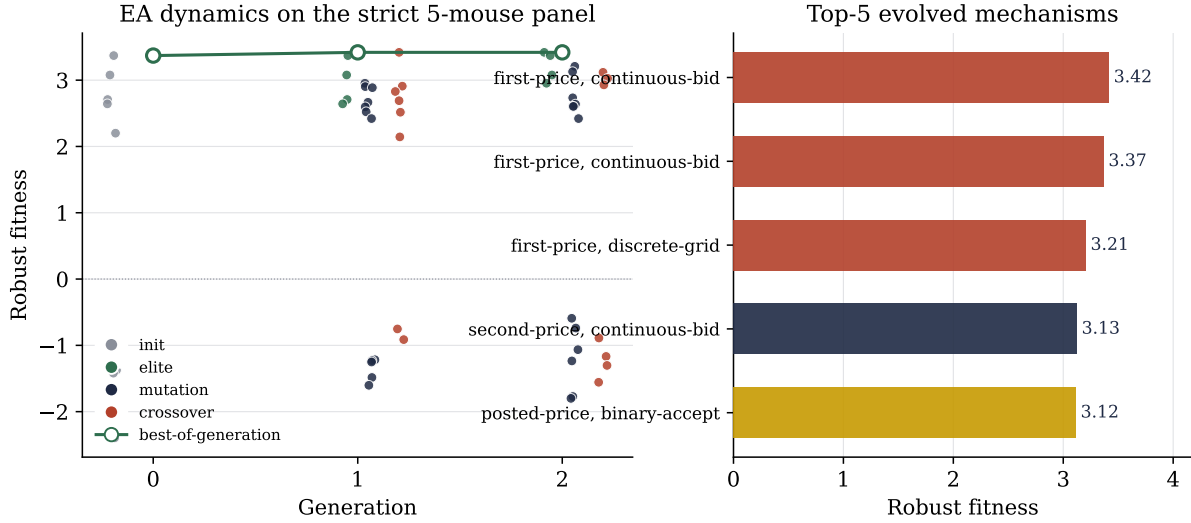


Figure 5: Evolutionary search over single-item auction genotypes against the strict 5-mouse panel (run 9: 24 individuals $\times$ 2 generations $\times$ 5 personas $\times$ 2 reps = 480 raw bidder-decisions). Left: robust fitness of each individual by generation, colored by operator; the green line traces the best-of-generation. Right: the top-five unique mechanisms, colored by payment rule.

We read this as *convergent validation* of two procedures rather than as a statement of EA optimality. The a-priori choice in Section 7 rested on sharp theory and human data; the EA choice rests on panel-based robustness. They agree, which they need not have done.

8.5 Caveats

Two generations is a proof-of-concept of the EA-plus-panel loop, not a paper-scale search; the convergence we report is qualitative agreement with Section 7 rather than a statement about optimality over the full genotype space. The fitness weights are the Stage-C-aligned defaults of Section 3.1 and have not been re-tuned to this specific search. Cross-model re-scoring of the survivors, along the lines of Section 7’s strict-panel portability check, is a natural extension and would let us flag candidates that scored well on the search model but degrade on a stronger held-out model.

Table 7: Top-five evolved mechanisms from run 9, ordered by robust fitness. Three of five are first-price sealed-bid; all five share no reserve and random tie-breaking. The gen-1 crossover child improves on every gen-0 initialization.

Rank	Payment rule	Bid language	Reserve	Framing scaffold	Fitness	First seen
1	first-price	continuous	none	none	3.42	gen 1, crossover
2	first-price	continuous	none	none	3.37	gen 0, init
3	first-price	discrete-grid	none	decision-table	3.21	gen 2, mutation
4	second-price	continuous	none	none	3.13	gen 2, mutation
5	posted-price	binary-accept	none	short-hint	3.12	gen 2, crossover

Notes. “Framing scaffold” is a presentation field of the genotype: **none** leaves the rule as a plain menu description; **decision-table** and **short-hint** are alternative scaffoldings the search may attach. No top-five mechanism uses a chain-of-thought demonstration scaffold; the genotype field is present and exploring it is left to a larger run.

9 Discussion

Procedural basis versus authored error mice. The construct-validity result is strongest for the procedural basis, where a single instruction (repair an axis) produces the predicted directional movement. The calibration result is more nuanced: the basis covers truthful and underbidding behavior on its own, but overbidding—a first-order regularity in human auction data (Kagel and Levin, 1993)—is recovered only with hand-specified heuristic strains. This distinction matters for interpretation. The model-agnostic claim applies cleanly to the procedural axes; the error mice are a complementary, more bespoke device for spanning the mistake space. The two should be reported as distinct epistemic objects.

Model dependence. Because deficits are defined as procedural controls rather than artifacts of one model, the protocol can be rerun on a different frontier model. At the strain level (Section 7), four of five reserved strict strains preserve their dominant behavioral class across base models, the lone exception being the contingent-reasoning deficit, which a stronger model tends to repair into truthful bidding. This is a feature of the design rather than a bug—the panel is precisely the instrument that makes such repair visible—but it cautions against reading single-model panel scores as model-invariant and identifies contingent reasoning as the deficit most in need of a hardened scaffold. The chain-of-thought result (Section 6) suggests one remedy: demonstration induces deficits more forcefully than instruction, and so should be harder for a capable model to override.

External validity. Digital mice are a phase-I screening population. A strain reproducing a mistake class establishes that the panel can express that behavior, not that the human share of that behavior is matched. We therefore keep the stress-testing claim (this paper) separate from any calibrated prediction of human behavior, which would require human pilot data.

10 Conclusion

We show that a frontier reasoning model can be instructed into stable, separable cognitive deficits whose behavior in a second-price sealed-bid auction (i) moves in the theory-predicted direction as each cognitive axis is repaired and (ii) spans the canonical space of human SPSB mistakes once a small set of targeted error strains is added, and (iii) reproduces sharp mechanism regularities in a first-price stress test—reasoning strains shade to the risk-neutral BNE and overbid relative to it, while an overgeneralizing strain bids above value and loses money. We further find that worked chain-of-thought demonstrations induce deficits more forcefully than instructions, that most reserved deficits preserve their behavioral class across base models, and that an evolutionary search driven by the panel as its fitness function recovers first-price sealed-bid as the top of the population, agreeing with the a-priori choice. This supports a research program in which instructed language-model agents are used as a phase-I population to screen mechanisms for robustness to bounded rationality before human-subject experiments are run. The immediate next steps are to extend the evolutionary, cross-model, and first-price analyses at scale and to carry both the construct-validation and calibration protocols into a second domain (matching).

References

- Crawford, V. P., and N. Iriberri (2007). “Level- k Auctions: Can a Nonequilibrium Model of Strategic Thinking Explain the Winner’s Curse and Overbidding in Private-Value Auctions?” *Econometrica*, 75(6), 1721–1770.
- Eyster, E., and M. Rabin (2005). “Cursed Equilibrium.” *Econometrica*, 73(5), 1623–1672.
- Kagel, J. H., and D. Levin (1993). “Independent Private Value Auctions: Bidder Behaviour in First-, Second- and Third-Price Auctions with Varying Numbers of Bidders.” *The Economic Journal*, 103(419), 868–879.
- Li, S. (2017). “Obviously Strategy-Proof Mechanisms.” *American Economic Review*, 107(11), 3257–3287.

Vickrey, W. (1961). “Counterspeculation, Auctions, and Competitive Sealed Tenders.” *The Journal of Finance*, 16(1), 8–37.